

Recommendations for Improving Clock Tree Synthesis QoR and Debugging QoR Issues

Agenda

- **Improving CTS QoR**
 - General Guidelines
 - Recommended Methodology (ICC-RM)
- **Debugging CTS QoR Issues**
 - Attributes on Clock Trees
 - Debugging CTS QoR Issues
 - Pre-CTS Analysis
 - Post-CTS Analysis
 - GUI Features

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General Guidelines

- Run the `check_clock_tree` command and address any issues identified by the tool
- Use buffers and inverters with a minimal difference between its rise and fall delay
- Use default values for CTS constraints:
 - Tightening `max_trans`, `max_cap`, `max_fanout` will degrade the clock tree power considerably
 - CTS derives the constraints from the technology data of the design. The derived constraints will give good insertion delay, skew, power, and area
 - To meet design requirement, If it is necessary to force CTS to use user specified constraints, set the `cts_force_user_constraints` variable to true

General Guidelines

- Avoid setting the `is_fixed` attribute on clock tree cells
 - The `is_fixed` attribute can cause CTS to take non-optimal path from insertion delay point of view
 - If necessary, specify only on a few cells
- Tightening the `max_buffer_levels` setting can result in huge DRC violations and poor CTS results
 - The option is hidden in 2010.12 release
 - Default changed to 200 instead of 50
 - From 2010.12 the option does not accept value less than 200
 - This is not a tree level constraint but a net level constraint

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Recommended Flow

```
source -e -v common_optimization_settings_icc.tcl
source -e -v common_placement_settings_icc.tcl

# CTS & CTO
source -e -v common_cts_settings_icc.tcl
set_delay_calculation -clock_arnoldi
check_clock_tree
clock_opt -only_cts -no_clock_route

# Timing optimization
source -echo common_post_cts_timing_settings.tcl
remove_ideal_network [all_fanout -flat -clock_tree]
clock_opt -no_clock_route -only_psyn \
    -area_recovery

# Clock routing and post route CTO
source -e -v common_route_si_settings_zrt_icc.tcl
route_zrt_group -all_clock_nets \
    -reuse_existing_global_route true
optimize_clock_tree -routed_clock_stage detail
```

```
reset_clock_tree_references

set_clock_tree_references -references
$ICC_CTS_REF_LIST

set_clock_tree_references -
delay_insertion_only -references
$ICC_CTS_REF_DEL_INS_ONLY

set_clock_tree_references -sizing_only -
references $ICC_CTS_REF_SIZING_ONLY

set_clock_tree_options -routing_rule NDR_2S \
    -use_default_routing_for_sinks 1

set_clock_tree_options -layer_list
$ICC_CTS_LAYER_LIST
```

```
set_app_var
timing_remove_clock_reconvergence_pessimism
true
...
```

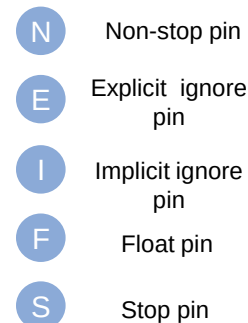
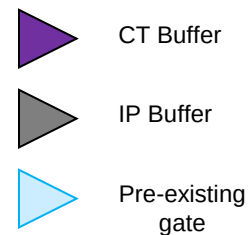
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Attributes on Clock Trees

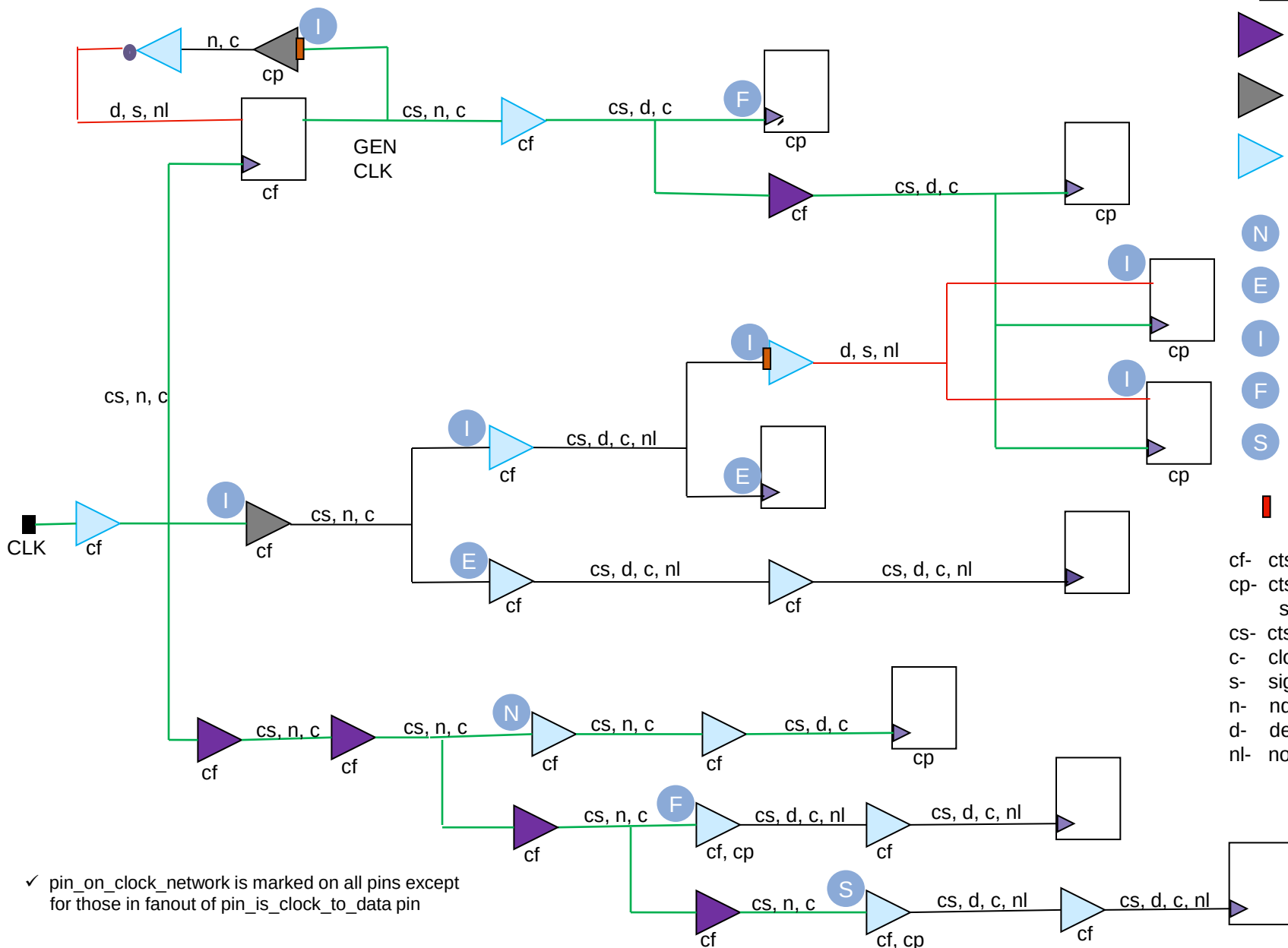
- IC Compiler sets attributes on clock tree elements for the following reasons:
 - To distinguish clock tree elements from data path elements
 - To prevent timing optimization from modifying the clock network
 - To guide extraction and routing
- In the following slide, attributes that are set on a clock tree under different situations are shown for a small design

Legend



pin is clock to data

cf- cts_fixed
 cp- cts_in_place_size_only
 cs- cts_synthesized
 c- clock net
 s- signal net
 n- ndr routing rule
 d- default routing rule
 nl- no layers assigned



✓ pin_on_clock_network is marked on all pins except for those in fanout of pin_is_clock_to_data pin

Attributes on Clock Tree

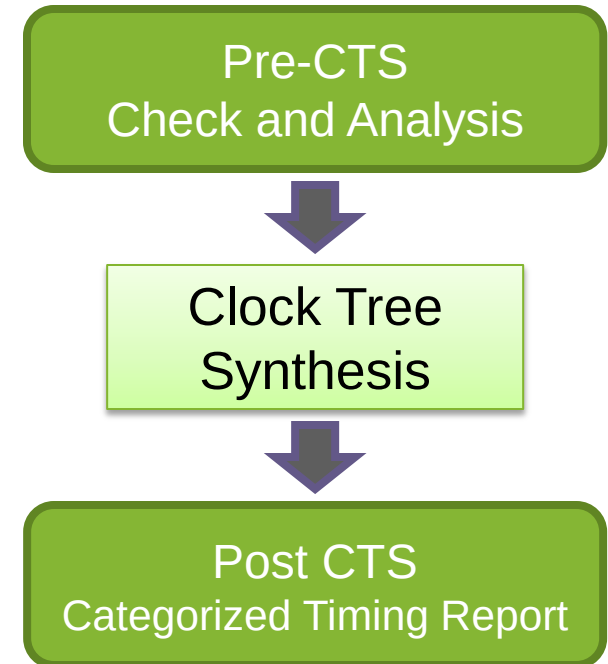
Attribute	Purpose
pin_on_clock_network	Set to <code>true</code> on pins of the clock network, to prevent timing optimization from touching these paths
pin_is_clock_to_data	Set to <code>true</code> on pins where the clock network stops and the data network begins. By default, CTS will not go beyond these pins. Timing optimizations can work on paths beyond such pins.
net_type	Set to <code>clock</code> on nets which are in connected to clock tree elements
cts_fixed	Set on to <code>true</code> on CTS inserted cells and pre-existing cells, to prevent them from being moved
cts_in_place_size_only	Set to <code>true</code> on sinks, and IP buffers which go to data pins. This attribute allows timing optimization to size them and prevents moving them.
cts_synthesized	Set to <code>true</code> on nets which are synthesized by CTS
Non-default rule	User specified non-default rules are set on clock nets which are in balanced portion of the clock tree

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Debugging CTS QoR Issues

- Challengers faced by designers:
 - Clock structure and constraints become more complicated
 - More chances of having inappropriate clock structure and constraints
 - Inappropriate clock structure & constraints lead to poor QoR
- Pre-CTS analysis will help prevent inappropriate inputs to CTS engine
 - `check_clock_tree`
 - GUI based analysis
- Post-CTS categorized report will help debug poor skew and insertion delay
 - `create_qor_snapshot -clock_tree`
 - Log file based debugging using Tcl script
 - GUI based analysis



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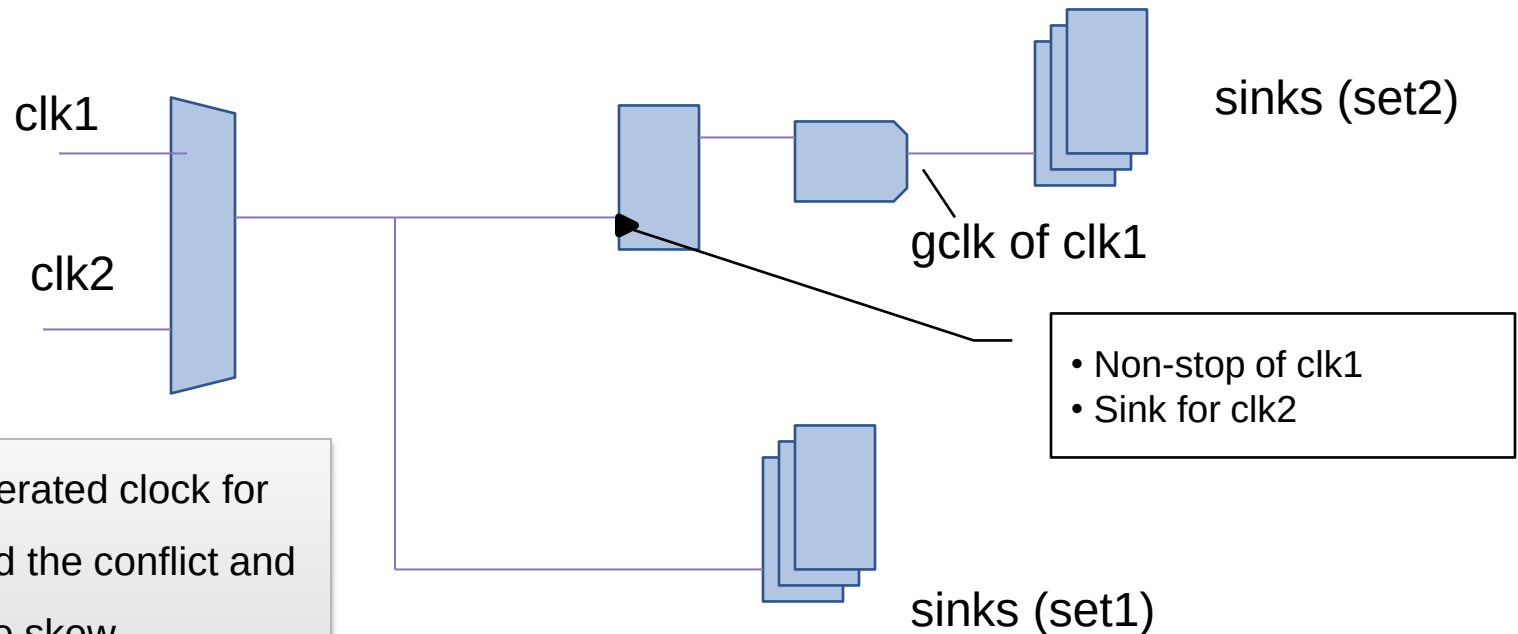
Pre-CTS Analysis:

`check_clock_tree`

- This command checks the following, and reports issues that could lead to bad QoR:
 - Clock tree structure
 - Constraints
 - Clock tree exceptions
- Below are few example issues that are reported:
 - Master clock not reaching generated clock
 - Conflicting balancing requirements at clock pins
 - Ignored exceptions along the clock tree path

Case Study 1: Large Skew for Clock Tree

- `check_clock_tree` reports CTS-872:
 - Shows pins which are treated differently for different clocks
- The clock structure is:



Creating generated clock for
clk2 resolved the conflict and
improved the skew

Case Study 2:

Unable to Meet Skew Target

- The specified skew target is 100ps, but tool gives 120ps
- `check_clock_tree` reports CTS-851:
 - Warning: There are multiple arcs between input and output pins for lib/AO16X. (CTS-851)

- Timing arcs:

Cell	Attributes	#	Sense/Type	From	To	When	Id
AO16X	u	0	pos unate	A	Z	B*C'*D	
		1	pos unate	A	Z	B*C'*D'	
		2	pos unate	A	Z	B*C*D'	
		3	pos unate	B	Z	A*C'*D	
		4	pos unate	B	Z	A*C'*D'	
		5	pos unate	B	Z	A*C*D'	

- Why is this an issue?
 - There is a delay difference of ~10ps between the arcs from B->Z
 - Longest path uses late delay (bigger of all) and Shortest path uses early delay (smallest of all)
 - Skew of 10ps from one cell!
 - There are 6 of these cells along the path contributing to ~60ps skew in total
- The clock path was going to output port (defined as stop pin)
 - Setting exclude on output port gives 85ps of skew!!

Case Study 3:

Poor Skew at Top Level

- Before performing CTS at top level, you need to ensure that the skew of all blocks (ILM or ETM) is good
- `check_clock_tree` reports CTS-871:
 - Warning: The phase delay skew of entrance pin blockA/buffer1/A in clock CLK is 0.500, larger than the threshold 0.050. (CTS-871)
- Why is this an issue?
 - For clock CLK, skew for the sub-tree inside the block is 500ps
 - It means the top level skew for CLK will be more than 500ps
- Analyze the block and identify the reason for poor skew at block level

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Before We Proceed...

- Main task of `compile_clock_tree`:
 - Build a DRC violation free clock tree
 - Perform skew improvement through gate hookup
 - Sets a clock path topology for `optimize_clock_tree` to work on
- Main task of `optimize_clock_tree`:
 - Perform clock net global routing
 - Perform skew and insertion delay optimization

CTS Log File Based Debugging With Tcl Script (SolvNet: 030705)

- Inputs required:
 - The log file generated by the `compile_clock_tree` command, with the `cts_use_debug_mode` variable set to true.
- Running the script :
 - Only IC Compiler shell is needed, design need not be loaded

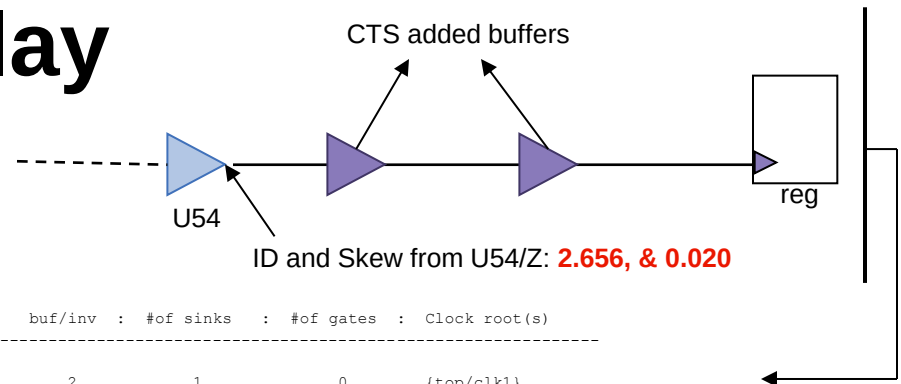
```
icc_shell> source debug_cts_from_log.tbc
icc_shell> debug_compile_clock_tree -cts_log_file cts.log
```
- The output report shows:
 - Gate-level driver pins where the skew and insertion delay are increased by more than threshold (default: 1ns for skew and 2ns for insertion delay)
 - Shows additional information such as number of sinks and gates connected to the driver pin, number of buffers inserted by Clock Tree Synthesis in that level, and the clock root to which this driver pin belongs
 - WHAT NEXT section to help identify clock nets which are bottlenecks for skew and insertion delay

Example 1: Large Insertion Delay

- Script report:

```
=====
ID jump Gates (sorted by ID jump):
=====
```

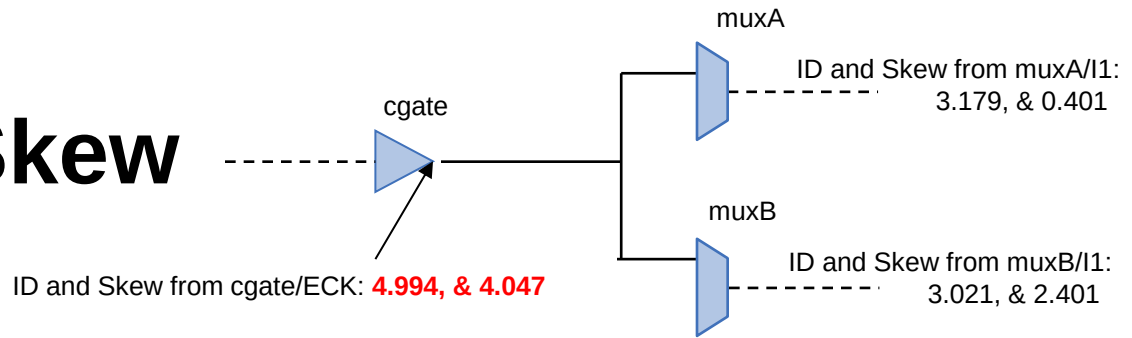
Gate pin	Gate level	Skew at gate	ID at gate	Skew jump	ID jump	buf/inv	#of sinks	#of gates	Clock root(s)
U54/Z	3	0.020	2.656	0.020	2.656	2	1	0	{top/clk1}
U63/Z	3	0.020	2.628	0.020	2.628	2	1	0	{top/clk2}



- Analysis of report:

- From the report, the gate U54 is at level 3, driving 1 sink and no other clock gates.
- Skew at the gate is 20ps and insertion delay at the gate is 2.656ns.
 - Jump in insertion delay is 2.656ns.
- Clock tree synthesis inserted 2 buffers resulting in a buffer tree of 2.656ns for one sink
 - Potentially a floorplan issue. You need to highlight this portion in the layout and check.
- The gate is in a clock tree whose root is top/clk1.
- Same issues exists with gate U63

Example 2: Poor Clock Skew



• Script report:

```
=====
Skew jump Gates (sorted by ID jump):
=====
Gate pin      : Gate level : Skew at gate : ID at gate : Skew jump : ID jump : buf/inv : #of sinks : #of gates : Clock root(s)
-----
cgate/ECK     15           4.047         4.994       1.646      1.815    0         0         2         clk
```

ID jump: $4.994 - 3.179 = 1.815$
 Skew jump: $4.047 - 2.401 = 1.646$

• Analysis of report:

- From the report, the gate **cgate** is at level 15, driving 0 sink and 2 other clock gates.
- Skew at the gate is 4.047ns and insertion delay at the gate is 4.4994ns.
 - Jump in skew is 1.646ns.
- Clock tree synthesis inserted no buffers on this net driven by **cgate/ECK**
 - There could be a `dont_buffer_net` attribute or a hierarchical `dont_touch` attribute set on the net or the net is inside an ILM.
 - Check the log file to find the actual cause.
- The gate is in a clock tree whose root is **clk**

Example3:

High Insertion Delay & Skew

Log excerpt from real design:

```
...
CTS: gate level 2 clock tree synthesis
CTS:   clock net   : clk1_net/z
...
Warning: Stop to insert buffers to fix DRC on net
        clk1_net/z (current buffer levels = 10, constraint =
        9). (CTS-233)
...

CTS: gate level 2 clock tree synthesis results
CTS:   clock net   : clk1_net/z
CTS:   driving pin: clk1_gate/z
CTS:   load pins   : 1 sink pins, 1 gates/macros pins, 3
        ignore pins
CTS:     buffer level 1: BUF4X (1)
...
CTS:     buffer level 10: BUF4X (1)
CTS:     clock tree skew      = worst[5.749]
CTS:     longest path delay   = worst[8.508](rise)
CTS:     shortest path delay  = worst[2.759](rise)
CTS:     total capacitance    = worst[1.555 1.555]
...
CTS:   gate/macro pin's phase delay
CTS:     1: worst[5.300](rise), worst[4.700](rise):
        clk1_gate2/ECK
...
```

→ User sets max_buffer_levels to 3 (default is 50 prior to 2010.12; 200 and hidden from 2010.12)

→ CTS-233 is shown during synthesis of clock net clk1_net/z.

→ This can result in poor skew and insertion delay, and DRC violations

→ Synthesis of this net results in insertion delay of 8.5ns and skew of 5.7ns

→ The skew & insertion delay at driven gate are OK.

→ Looking at the synthesis log of this gate level, we see CTS-833.

→ Allow CTS to use the default value for max_buffer_levels option.

Post-CTS Analysis:

CTS Categorized Timing Report

- CTS categorized timing report will analyze the clock tree paths for potential issues, then summarize the findings in an html report
- For each clock in the design, the following is analyzed:
 - Longest delay path
 - Shortest delay path
 - Skew bottleneck pin list
- Usage:

```
create_qor_snapshot -clock_tree  
query_qor_snapshot -name
```

CTS Categorized Timing Report

1. GUI Interface - Interactive Report

Query QoR Snapshot

Snapshots
Snapshot input directory:
Snapshot name:
Query mode: ☒ Interactive Report ☐ Case sensitive

CTR Interactive Viewer

File:

[Create/Remove Exceptions](#)

Highlighted Objects in Layout Window

☒ Longest ☒ Shortest ☒ Intrinsic ☒ Buffers ☒ Gates
☐ Transition Deg.

Input file: /global/wwas_northeast9/dpower/2010.12/coreblk_mcmmy/snapshot/postcts.tim.max.rpt
Filters: -transition_violation {1} -fanout_violation {1} -capacitance_violation {1} -dont_touch {1} -dont_buffer {1} -dont_size {1} -size_only {1} -explicit_ignore {1} -skew {0.050} -transition_deg {2.0} -slew_deg {2.0}
And Columns: none
Sort Column: skew (descending)
Number of Paths: 2

Exception	Display	Clock	Clock Tree Name	Scenario	Pathname	Master Clock	Master Source	Total Sink Pins	Global Skew	Delay	Trans. Viol.	Cap. Viol.	Explicit Ignore Pins	Dont Touch Subtree Pins	Dont Buffer Nets	Dont Size Cells
±	☐	☐	C1	tck	default	All	-	-	370	0.17175	1.3870	0	0	0	0	0
±	☐	☐	C0	clk	default	All	-	-	22335	0.04700	2.4510	0	1	0	0	0

CTS Categorized Timing Report

2. GUI Interface - Cross-Reference to Clock Report

report_clock_tree

```
Report : clock tree
Design : coreblk
Version: E-2010.12-ICC-SP1
Date   : Fri Feb 25 08:22:38 2011
*****

Information: Float pin scale factor for the 'max' operating condition

===== Global Skew Report =====

Clock Tree Name      : "clk"
Clock Period         : 3.30000
Clock Tree root pin  : "clk"
Number of Levels     : 27
Number of Sinks      : 22373
Number of CT Buffers : 254
Number of CTS added gates : 0
Number of Preexisting Gates : 747
Number of Preexisting Buf/Inv : 15
Total Number of Clock Cells : 1013
Total Area of CT Buffers : 1209.95813
Total Area of CT cells : 8314.58984
Max Global Skew      : 0.71456
Number of MaxTran Violators : 2
Number of MaxCap Violators : 0
```

report_clock_tree - exceptions

```
(J) = Implicit nonstop pin
(T) = Explicit nonstop pin
(D) = Dont touch subtree pin
(B) = Dont buffer net
(S) = Dont size cell

(O) = Size only cell

1. Clock: clk
Clock root: clk
Clock net: clk
Total sinks: 22373
Explicit ignore pins: 0
Explicit sync pins: 4
(F) pwd/iram/iram_array/clk ( 1.800 1.800 1.800 1.800 )
(F) pwd/lmcacheram/lm_1/clock ( 1.000 1.000 1.000 1.000 )
(F) pwd/lmcacheram/lm_0/clock ( 1.000 1.000 1.000 1.000 )
(F) pwd/iccacheram/icarr/clock ( 1.200 1.200 1.200 1.200 )
Implicit ignore pins: 106
(I) pd_iso/pd/ut/rf/gcgen_wb/g/u5_gtd_clk/CP
(I) pd_iso/pd/ut/rf/gcgen_wb/g/CKND8HVT_G1B2I2_1/I
(I) pd_iso/pd/ut/mul2/muocg/u5_gtd_clk/CP
(I) pd_iso/pd/ut/preg_outog/u5_gtd_clk/CP
(I) pd_iso/pd/ut/rf/imm_g1f_reg_state_q_reg_7_/latch/CP
(I) pd_iso/pd/ut/rf/imm_g1f_reg_state_q_reg_15_/latch/CP
```

And Columns: none
Sort Column: skew (descending)
Number of Paths: 2

Exception	Display	Clock	Clock	Clock	Scenario	Pathname	Master	Master	Total	Global	De	Trans.	Cap.	#	Worst	Implicit	Explicit
							Clock	Source	Sink	Skew		Viol.	Viol.	Levels	Incr	Ignore	Sync Pins
									Pins					of			
														Logic			
±			C0	clk	default	All	-	-	22373	0.71456	3.0020	2	0	27	0.4670	106	4
±			C1	tck	default	All	-	-	370	0.14743	1.5110	0	0	26	0.2190	0	0

CTS Categorized Timing Report

3. GUI Interface - Cross-Reference to Layout

The screenshot displays the IC Compiler GUI with the 'LayoutWindow.1' window active. The 'Block Implementation' tab is selected, showing a detailed layout of the circuit. A 'CategorizedClockReport' window is open on the right, displaying timing data for various clock signals. A purple arrow points from the 'Clock Tree Name' column in the report to the corresponding clock signal in the layout.

Highlighted Objects in Layout

☒ Longest ☒ Shortest
☐ Trans. Viol. ☐ Transition Deg.

Input
file: /remote/testcases24/9000176001_9
Filters: -transition_violation {1} -fanout_violation {1} -size_only {1} -explicit_ignore {1} -skip {1}
And Columns: none
Sort Column: logic_levels (ascending)
Number of Paths: 5

Exception Display Clock

Exception	Display	Clock	Clock Tree Name	Scenario
±	☐	☐	clk	func_min
±	☐	☒	C0	pci_clk
±	☐	☐	C3	leon_clk
±	☐	☐	C4	test_clk
±	☐	☐	C1	leon_clk

CategorizedClockReport

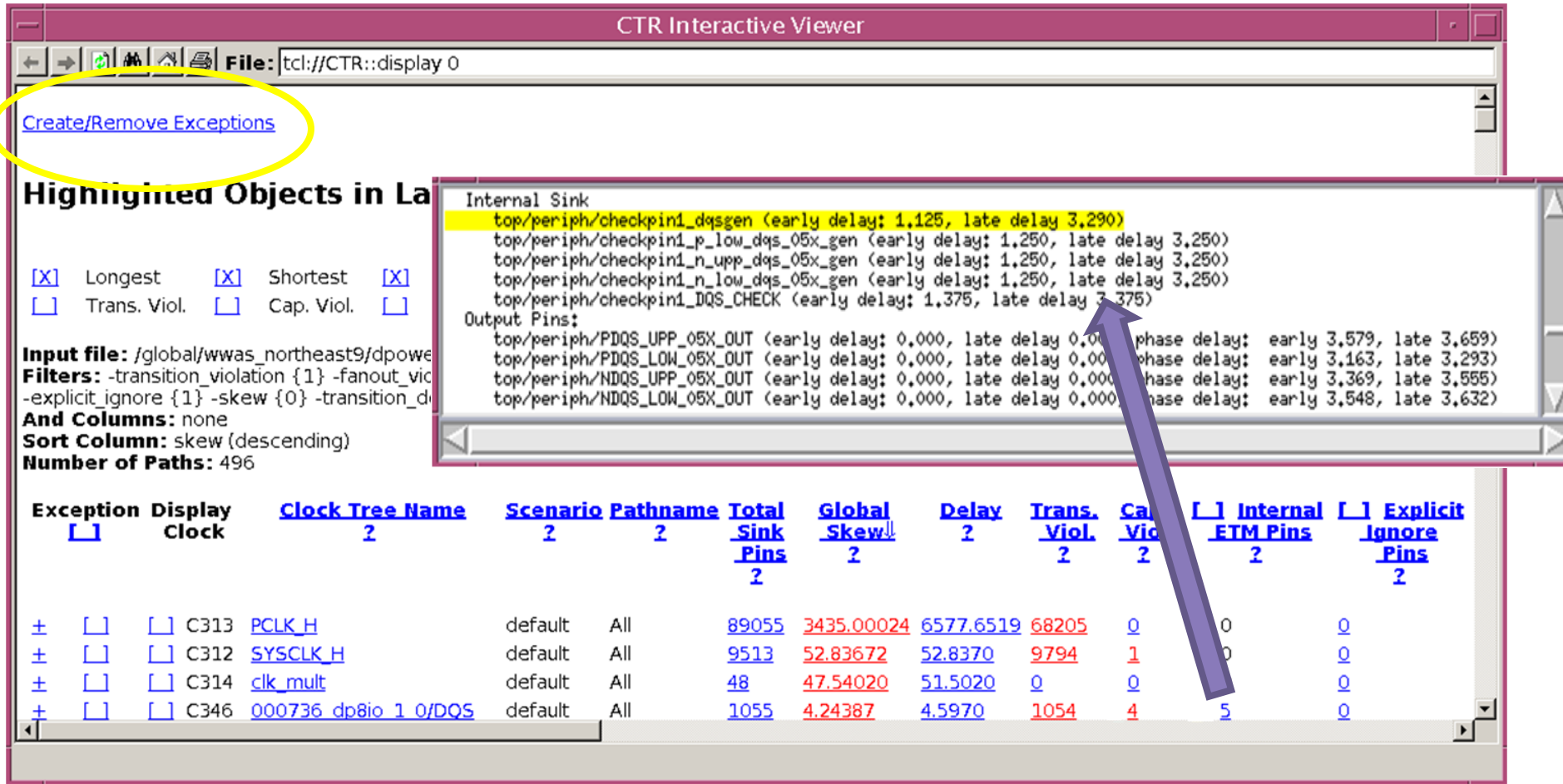
Apply Reload Options:

- ☒ p0 CTS Gates 5
- ☒ p0 CTS Buffers 8
- ☒ C0 Shortest Path 30
- ☒ C0 Longest Path 24

Clock	Logic Levels	Shortest Path	Longest Path	Setup	Hold	Recovery	Removal	Setup	Hold	Recovery	Removal
clk	5695	0.09776	0.5840	0	0	0	0	0	0	0	0
C0	5896	0.12315	0.8290	0	0	0	0	0	0	0	0
C1	5695	0.21527	1.3950	2	0	0	0	0	0	0	0

Post-CTS Categorized Timing Report

4. GUI Interface – Creating Exceptions



The screenshot shows the CTR Interactive Viewer window. The 'File' menu is open, and the 'Create/Remove Exceptions' option is highlighted with a yellow circle. A purple arrow points from this menu item to a detailed view of a specific exception.

Highlighted Objects in List:

- Internal Sink
 - top/periph/checkpin1_dqsgen (early delay: 1.125, late delay 3.290)
 - top/periph/checkpin1_p_low_dqs_05x_gen (early delay: 1.250, late delay 3.250)
 - top/periph/checkpin1_n_upp_dqs_05x_gen (early delay: 1.250, late delay 3.250)
 - top/periph/checkpin1_n_low_dqs_05x_gen (early delay: 1.250, late delay 3.250)
 - top/periph/checkpin1_DQS_CHECK (early delay: 1.375, late delay 3.375)
- Output Pins:
 - top/periph/PDQS_UPP_05X_OUT (early delay: 0.000, late delay 0.000, phase delay: early 3.579, late 3.659)
 - top/periph/PDQS_LOW_05X_OUT (early delay: 0.000, late delay 0.000, phase delay: early 3.163, late 3.293)
 - top/periph/NDQS_UPP_05X_OUT (early delay: 0.000, late delay 0.000, phase delay: early 3.369, late 3.555)
 - top/periph/NDQS_LOW_05X_OUT (early delay: 0.000, late delay 0.000, phase delay: early 3.548, late 3.632)

Input file: /global/wwas_northeast9/dpower
Filters: -transition_violation {1} -fanout_violation {1} -explicit_ignore {1} -skew {0} -transition_delay {0}
And Columns: none
Sort Column: skew (descending)
Number of Paths: 496

Exception	Display	Clock	Clock Tree Name	Scenario	Pathname	Total Sink Pins	Global Skew	Delay	Trans. Viol.	Cap. Viol.	Internal ETM Pins	Explicit Ignore Pins
±	☐	☐	C313 PCLK_H	default	All	89055	3435.00024	6577.6519	68205	0	0	0
±	☐	☐	C312 SYSCLK_H	default	All	9513	52.83672	52.8370	9794	1	0	0
±	☐	☐	C314 clk_mult	default	All	48	47.54020	51.5020	0	0	0	0
±	☐	☐	C346 000736 dp8io_1 0/DQS	default	All	1055	4.24387	4.5970	1054	4	5	0

Post-CTS Categorized Timing Report

Case Study: Huge Skew After CTO

CTR Interactive Viewer

File: tcl://CTR::display 0

[Create/Remove Exceptions](#)

Highlighted Objects in Layout Window

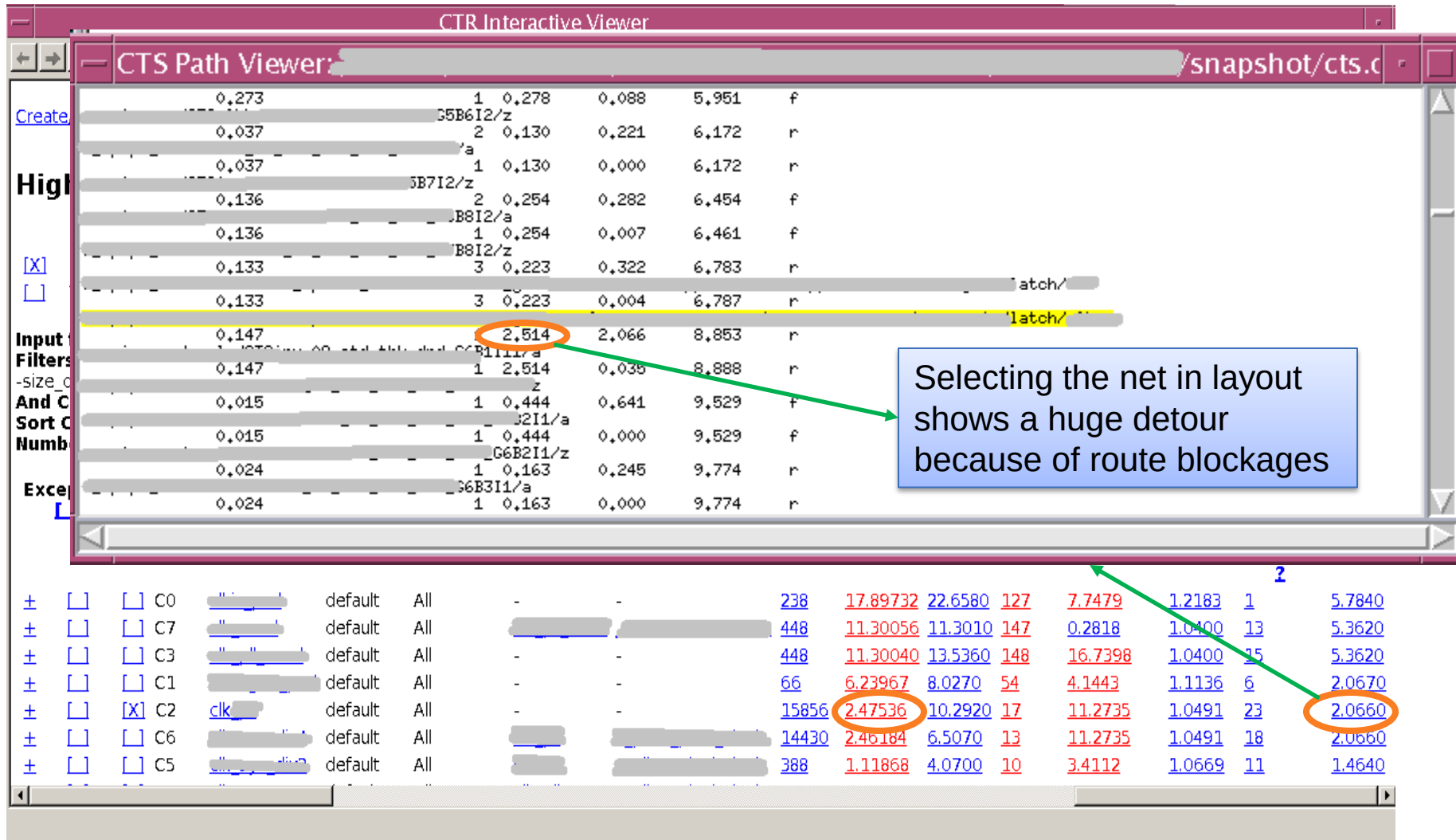
☒ Longest ☒ Shortest ☒ Intrinsic ☒ Buffers ☒ Gates
☐ Trans. Viol. ☐ Cap. Viol. ☐ Transition Deg.

Input file: /.../snapshot/cts.tim.max.rpt
Filters: -transition_violation {1} -fanout_violation {1} -capacitance_violation {1} -dont_touch {1} -dont_buffer {1} -dont_size {1}
 -size_only {1} -explicit_ignore {1} -skew {0} -transition_deg {2.0} -slew_deg {2.0}
And Columns: none
Sort Column: skew (descending)
Number of Paths: 12

Exception	Display	Clock	Clock Tree Name	Scenario	Pathname	Master Clock	Master Source	Total Sink Pins	Global Skew	Delay	Trans Viol	Transition Deg.	Slew Deg.	# Levels of Logic	Worst Incr
±	☐	☐	C0	default	All	-	-	238	17.89732	22.6580	127	7.7479	1.2183	1	5.7840
±	☐	☐	C7	default	All	-	-	448	11.30056	11.3010	147	0.2818	1.0400	13	5.3620
±	☐	☐	C3	default	All	-	-	448	11.30040	13.5360	148	16.7398	1.0400	15	5.3620
±	☐	☐	C1	default	All	-	-	66	6.23967	8.0270	54	4.1443	1.1136	6	2.0670
±	☐	☒	C2	default	All	-	-	15856	2.47536	10.2920	17	11.2735	1.0491	23	2.0660
±	☐	☐	C6	default	All	-	-	14430	2.46184	6.5070	13	11.2735	1.0491	18	2.0660
±	☐	☐	C5	default	All	-	-	388	1.11868	4.0700	10	3.4112	1.0669	11	1.4640

Post-CTS Categorized Timing Report

Case Study: Huge Skew after CTO



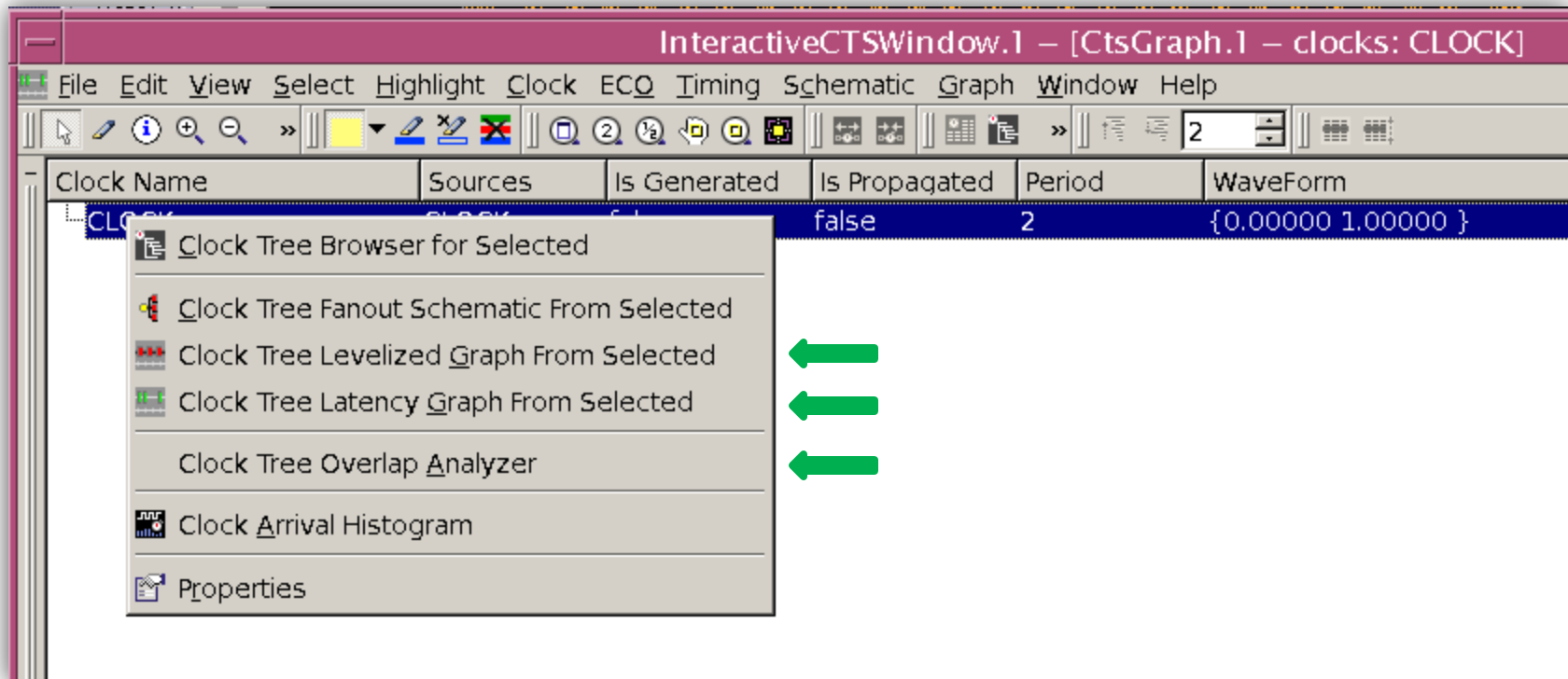
Agenda

- Improving CTS QoR
 - General Guidelines
 - Recommended Methodology (ICC-RM)
- **Debugging CTS QoR Issues**
 - Attributes on Clock Trees
 - Debugging CTS QoR Issues
 - Pre-CTS Analysis
 - Post-CTS Analysis
 - GUI Features

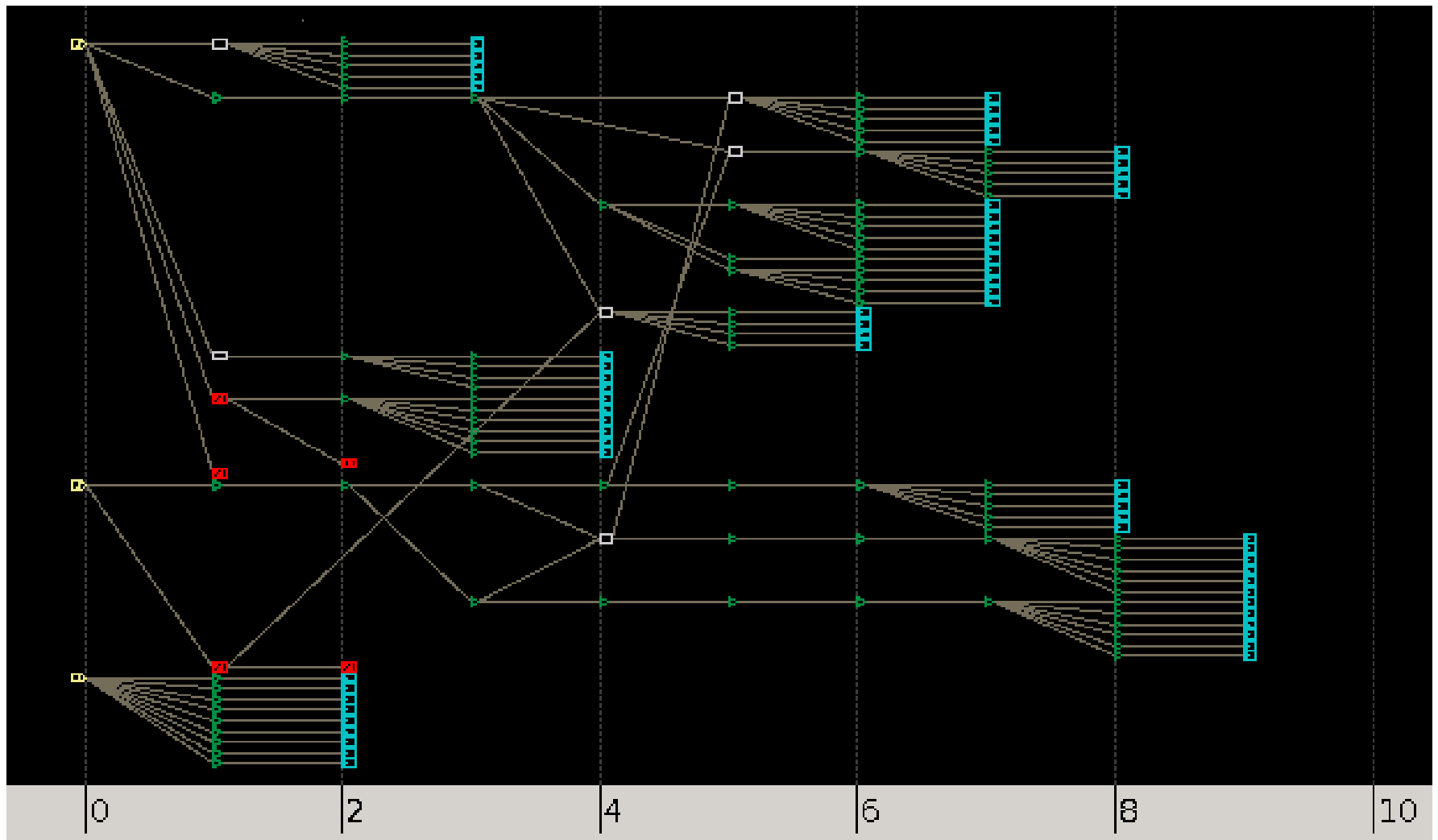
Clock Tree Analysis: GUI

- Three ways to analyze using GUI:
 - Levelized graph (pre-CTS, post-CTS)
 - Overlap analyzer (pre-CTS, post-CTS)
 - Latency graph (post-CTS)
- From 2010.12, multiple clock trees can be seen in one graph window:
 - Available for:
 - Clock levels graph
 - Clock latency graph
 - Easy to analyze interactions and interclock relationships between clocks

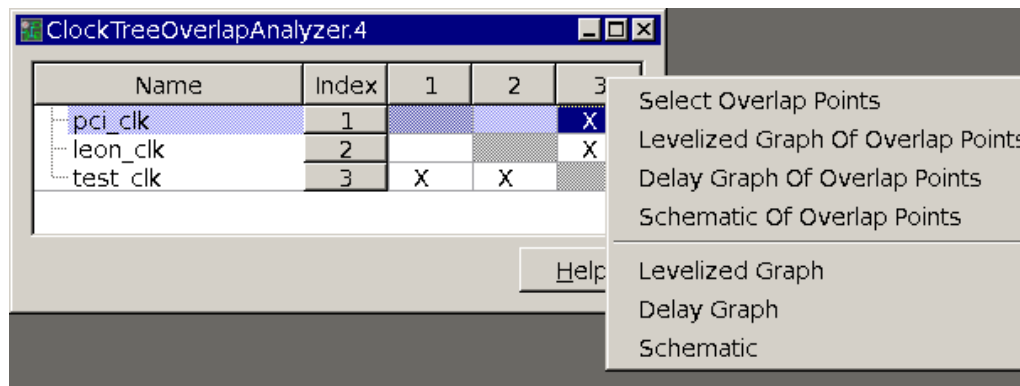
User Interface for Clock Tree Analysis



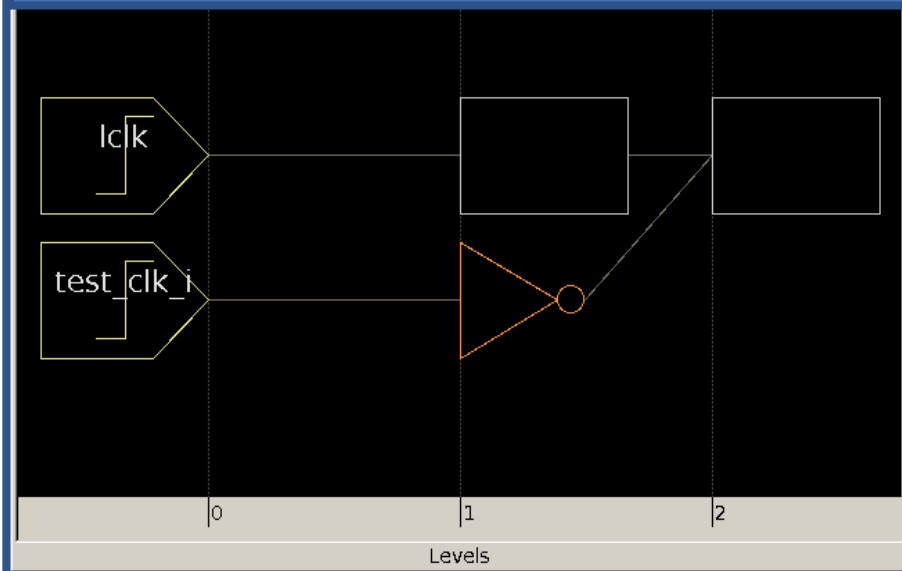
Multiple Tree Levelized Graph View



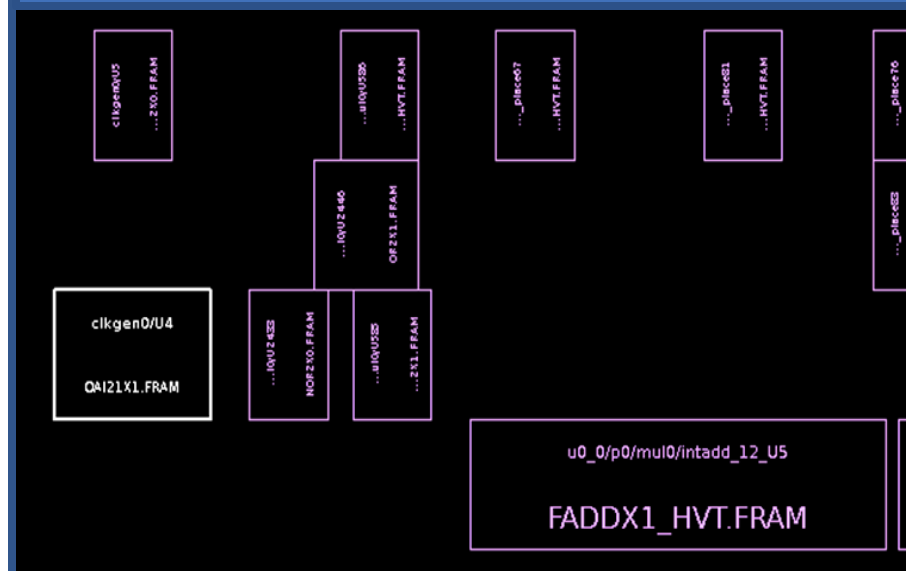
Multiple Clock Graph Overlap Analyzer



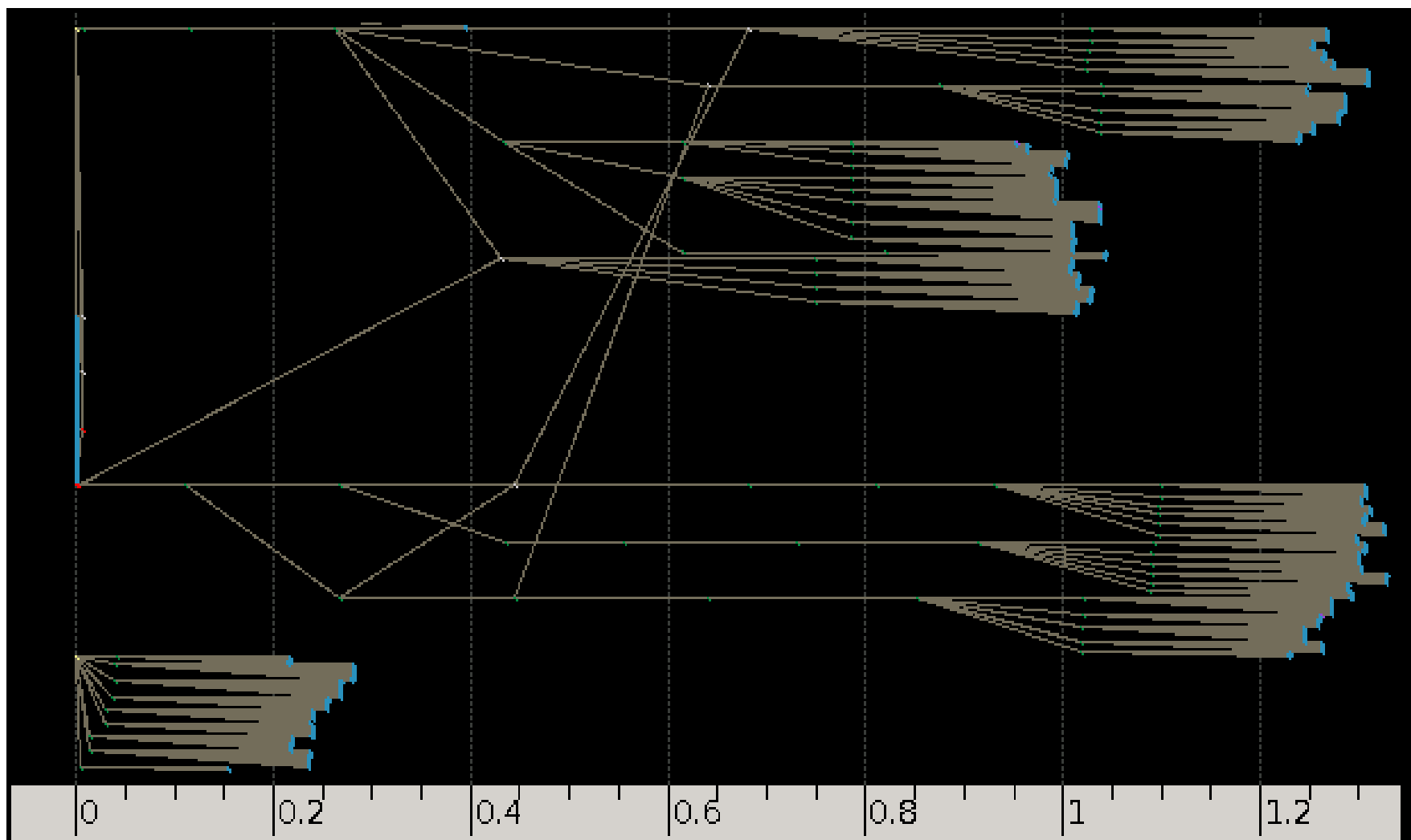
Levelized Graph of overlap point



Select overlap point in layout



Multiple Tree Latency Graph View

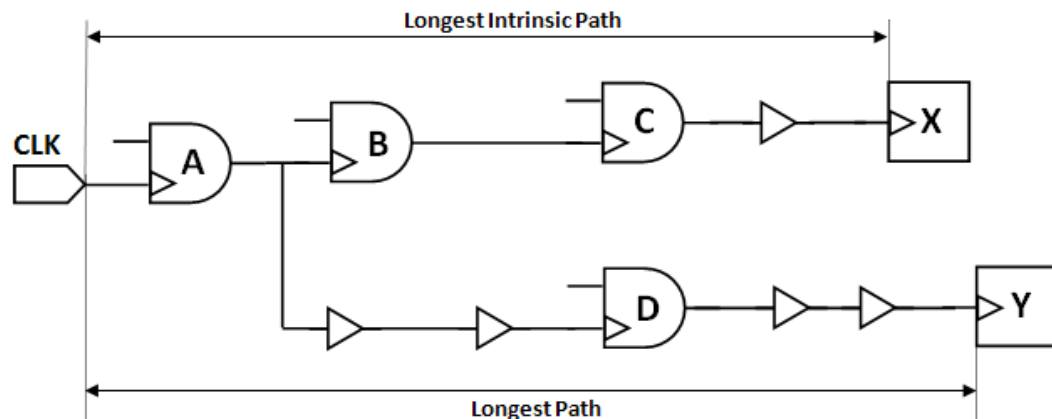


Clock Tree Latency Graph: Longest Intrinsic Path Delay

- Clock tree latency graph view
 - Provides a clock latency view of a selected clock network
 - The clock-graph view makes it easier to identify the worst intrinsic path
- Longest Intrinsic Path:
 - The longest clock path when only pre-existing cells delays are considered

Generally, the insertion delay of clock tree is controlled by longest intrinsic path

Can be helpful in identifying insertion delay bottlenecks



Latency Graph View: Example

